

Project Title: System Verification and Validation
Plan for CEWS System

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February 18, 2023

1 Revision History

Date	Version	Notes
2023-02-10	0.0	Creation
2023-02-14	1.0	Version 1.0

Contents

1	Revision History	i
2	Symbols, Abbreviations and Acronyms	iv
3	General Information	1
3.1	Summary	1
3.2	Objectives	1
3.3	Relevant Documentation	1
4	Plan	1
4.1	Verification and Validation Team	1
4.2	SRS Verification Plan	1
4.3	Design Verification Plan	2
4.4	Verification and Validation Plan Verification Plan	2
4.5	Implementation Verification Plan	2
4.6	Automated Testing and Verification Tools	2
4.7	Software Validation Plan	2
5	System Test Description	2
5.1	Tests for Functional Requirements	2
5.1.1	Area of Testing 1: FR1, FR4	2
5.1.2	Area of Testing 2: FR2, FR3, FR4	4
5.2	Tests for Nonfunctional Requirements	5
5.2.1	Area of Testing 1: NFR1	5
5.2.2	Area of Testing 2: NFR2	6
5.2.3	Area of Testing 3: NFR3	6
5.2.4	Area of Testing 4: NFR4	6
5.3	Traceability Between Test Cases and Requirements	7
6	Unit Test Description	7
	References	8
7	Appendix	9
7.1	Symbolic Parameters	9

List of Tables

1	CEWS Result on Benchmark Datasets from [1]	5
2	Which test cases are supporting which requirements	7
3	Table of SYMBOLIC.CONSTANTS	9

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test
CEWS	Cut Edge Weight Statistic
J_N	Normalized Multiclass Cut Edge Weight Measurement
D	Input Data Matrix
C	Input Class Matrix
FR	Functional Requirement
NFR	Non Functional Requirement

This document outlines the verification and validation plan for testing the CEWS System.

3 General Information

3.1 Summary

This document contains the verification and validation plan for CEWS System, a program for calculating the CEWS measurement for a dataset as shown in [1]. CEWS System takes a set of data points D and a list of class labels C as inputs and computes the Normalized Multiclass Cut Edge Weight Measurement J_N .

3.2 Objectives

The objective of this document is to build confidence in the software's correctness. This document provides the test plan used to verify and validate the final product. Therefore, it should aim to encompass all validation and verification components required for testing.

3.3 Relevant Documentation

Relevant documentation includes the system requirements specification found here: [GitHub Link](#).

4 Plan

4.1 Verification and Validation Team

The verification and validation team includes the domain expert Joachim de Fourestier, the verification and validation reviewer Mina Mahdipour, and Dr. Spencer Smith.

4.2 SRS Verification Plan

The SRS will be reviewed by the SRS review team, consisting of the domain expert Joachim de Fourestier and the SRS reviewer, Sam Joseph Crawford, and Dr. Spencer Smith.

4.3 Design Verification Plan

The design document MIS will be reviewed by the MIS review team, consisting of the domain expert Joachim de Fourestier and the MIS reviewer, Deesha Vipulkumar Patel, and Dr. Spencer Smith.

4.4 Verification and Validation Plan Verification Plan

This document will be reviewed by the VnV team members in Section [4.1](#).

4.5 Implementation Verification Plan

Individual tests are outlined in sections 5.1 through 5.2. The tests will either be performed manually or with someone reviewing the code.

4.6 Automated Testing and Verification Tools

The VnV process will use no tools.

4.7 Software Validation Plan

To test the output of the software, results from the paper [\[1\]](#) will be used as a pseudo-oracle. The following benchmark datasets from the UCI Machine Learning Repository will be used: Breast Cancer, BUPA liver, Glass Ident, Haberman, Image seg, Ionosphere, Iris (Bezdek), Iris plants, Musk "Clean1", Pima Indians, Waveform, Wine recognition and Yeast.

5 System Test Description

5.1 Tests for Functional Requirements

5.1.1 Area of Testing 1: FR1, FR4

This area of testing will address inputs to the program.

Functional requirement 1 states: CEWS System will read the input data. Also functional requirement 4 states: The CEWS System will generate an output.

Tests for FR1

1. T1 - Normal Input

Control: Manual

Initial State: Pending Input

Input: D and C have the same number of samples and features, C contains at least two classes.

Output: No error generated, program proceeds to next state.

Test Case Derivation: FR1, FR4

How test will be performed: Manual

2. T2 - Data Mismatch

Control: Manual

Initial State:

Input: D and C have different number of samples or D has a different number of features.

Output: Generate error about incorrect data format.

Test Case Derivation: FR1, FR4

How test will be performed: Manual

3. T3 - Single Class

Control: Manual

Initial State:

Input: C contains only one class

Output: Generate error about lack of classes.

Test Case Derivation: FR1, FR4

How test will be performed: Manual

4. T4 - Empty Data

Control: Manual

Initial State: Pending Input

Input: C or D is null.

Output: Generate warning about empty data object.

Test Case Derivation: FR1, FR4

How test will be performed: Manual

5. T5 - Value out of range

Control: Manual

Initial State: Pending Input

Input: C contains a value larger than $W_{\max}$ or smaller than $W_{\min}$. D contains a value larger than $F_{\max}$ or smaller than $F_{\min}$. (Table 3)

Output: Generate warning about data value out of range.

Test Case Derivation: FR1, FR4

How test will be performed: Manual

5.1.2 Area of Testing 2: FR2, FR3, FR4

This area of testing addresses the result of the program. Functional Requirement 2 states: CEWS System will preform the calculation.

Functional Requirement 3 states: The CEWS System should generate an error if the calculation result exceeds the constraints.

Functional Requirement 4 states: The CEWS System will generate an output.

1. T6 - UCI Dataset Tests

Control: Manual

Initial State: Pending Input

Input: UCI Machine Learning Repository datasets: Breast Cancer, BUPA liver, Glass Ident, Haberman, Image seg, Ionosphere, Iris (Bezdek), Iris plants, Musk "Clean1", Pima Indians, Waveform, Wine recognition and Yeast.

Output: Normalized CEWS Measurement J_N .

Test Case Derivation: FR2, FR3, FR4

How test will be performed:

- (a) Format a dataset
- (b) Input the reformatted dataset to the system
- (c) Check the output value against the results in Table 1.

Dataset	Expected Result
Breast Cancer	0.008
BUPA liver	0.401
Glass Ident.	0.356
Haberman	0.331
Image seg.	0.224
Ionosphere	0.137
Iris (Bezdek)	0.090
Iris Plants	0.087
Musk "Clean1"	0.167
Pima Indians	0.310
Waveform	0.255
Wine recognition	0.093
Yeast	0.524

Table 1: CEWS Result on Benchmark Datasets from [1]

5.2 Tests for Nonfunctional Requirements

5.2.1 Area of Testing 1: NFR1

Non Functional Requirement 1 states: The accuracy of the computed solutions should meet the level needed for data analysis. The absolute error should be less than ACCURACY_ERR_ABS.

1. T7: Accuracy

Type: Manual

Initial State: Pending input

Input/Condition: UCI Datasets listed in Table 1.

Output/Result: Less than ACCURACY_ERR_ABS difference in results from Table 1.

How test will be performed: Manual

5.2.2 Area of Testing 2: NFR2

Non Functional Requirement 2 states: A program should be able to call CEWS System by calling a function of the form $\text{CEWS}(D, C)$ and CEWS System should return the output as described in FR4.

1. T8: Usability

Type: Manual

Initial State: Off

Input/Condition: Simulated dataset

Output/Result: J_N

How test will be performed: Manually, the user will attempt to call the program's function.

5.2.3 Area of Testing 3: NFR3

Non Functional Requirement 3 states: The program code should be readable and clear.

1. T9: Maintainability

Type: Manual

Initial State: None

Input/Condition: None

Output/Result: None

How test will be performed: The author will review the code.

5.2.4 Area of Testing 4: NFR4

Non Functional Requirement 4 states: The software should run on the Windows 11 operating system

1. T9: Maintainability

Type: Manual

Initial State: Off

Input/Condition: Simulated dataset

Output/Result: J_N

How test will be performed: Manually on a computer running Windows 11.

5.3 Traceability Between Test Cases and Requirements

	T1	T2	T3	T4	T5	T6	T7	T8	T9
FR1	X	X	X	X	X				
FR2					X				
FR3					X				
FR4		X	X	X	X				
NFR1						X			
NFR2							X		
NFR3								X	
NFR4									X

Table 2: Which test cases are supporting which requirements

6 Unit Test Description

This section is intentionally left blank until the MIS is completed.

References

- [1] D. A. Zighed, S. Lallich, and F. Muhlenbach, “A statistical approach to class separability,” *Applied Stochastic Models in Business and Industry*, vol. 21, no. 2, pp. 187–197, Mar. 2005. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/asmb.532>

7 Appendix

This is where you can place additional information.

7.1 Symbolic Parameters

The definition of the test cases will call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance.

Table 3: Table of SYMBOLIC_CONSTANTS

Var	Value
$F_{\min}$	$-(2 - 2^{-23})2^{127}$
$F_{\max}$	$(2 - 2^{-23})2^{127}$
$W_{\min}$	0
$W_{\max}$	$2^8 - 1$
$J_{\max}$	1
$J_{\min}$	0
ACCURACY_ERR_ABS	0.01